8.4 System Administration

As a developer, you're of course primarily responsible for coding. In real life, however, the clear distinction between administration and software development sometimes becomes blurred. You need to set up Docker containers, network setups, virtual machines, or database servers (even if only as a test environment). As the following examples show, AI tools can support you with clear instructions.

We didn't want to call this section "Linux System Administration," but we have to admit that the following pages primarily deal with Linux. This doesn't (only) have to do with personal preference, but mainly with the fact that Linux system administration is done in text mode. You run commands in the terminal, change configuration files, and so on. A language model can be a good help here.

In contrast, Windows administration is often carried out in graphical user interfaces. This doesn't suit the nature of language models. In addition, as of early 2025, ChatGPT and others aren't yet able to support explanations with screenshots of dialogs or setup wizards.

8.4.1 SSH Authentication by Key

The first example is about automatically copying files to a Linux server via `scp`. Secure Shell (SSH) keys must be used to avoid having to enter a password each time a script is executed.

Prompt: I want to use `scp` to back up files to a Linux server. The script should run without interaction. What's a safe way to accomplish this?

> To use `scp` for backing up files to a Linux server without requiring interaction, you can use key-based authentication. Here's a step-by-step guide to set this up.

> - First, generate an SSH key pair on your local machine:
>
>   ```
>   ssh-keygen -t rsa -b 4096 -C "your_email@example.com"
>   ```
>
>   Follow the prompts. When asked for a passphrase, you can choose to leave it empty for automated processes, but it's recommended to use a passphrase for added security. The key pair is typically saved in `~/.ssh/id_rsa` and `~/.ssh/id_rsa.pub`.
> - Use `ssh-copy-id` to copy your public key to the remote server:
>
>   ```
>   ssh-copy-id username@remote_server_ip
>   ```
>
>   This command will prompt for your password on the remote server. After successful execution, your public key will be added to the `~/.ssh/authorized_keys` file on the remote server.
> - Verify that you can connect to the remote server without being prompted for a password:
>
>   ```
>   ssh username@remote_server_ip
>   ```

ChatGPT then provides an example of a simple backup script and its automation via Cron. We've deleted this part of the output here.

Basically, SSH key authentication also works if the script is supposed to run on Windows. However, the setup is a little more complicated. ChatGPT explains how to manually set up the public part of the key on the server.

Prompt: My backup script should run on Windows. However, `ssh-copy-id` isn't available. Can I work around this problem?

> Yes, you can work around this issue on Windows by manually copying your SSH public key to the remote server. First, generate an SSH key pair using the built-in OpenSSH client in Windows 10/11.
>
> ```
> ssh-keygen -t rsa -b 4096 -C "your_email@example.com"
> ```

> Follow the prompts to save the key (by default, it will be saved to `C:\Users\your_username\.ssh\id_rsa` and `C:\Users\your_username\.ssh\id_rsa.pub`).
>
> Since `ssh-copy-id` isn't available on Windows, you need to manually copy the contents of the public key file (`id_rsa.pub`) to the `~/.ssh/authorized_keys` file on the remote server. Open the `id_rsa.pub` file in a text editor and copy its contents. Connect to your remote server using an SSH client. Once logged in, open or create the `~/.ssh/authorized_keys` file on the remote server:
>
> ```
> nano ~/.ssh/authorized_keys
> ```
>
> Paste the copied public key into the `authorized_keys` file and save it.

8.4.2 Docker Setup

To test a database client, you need a local MySQL server. You don't want to perform a "real" installation, but want to use Docker. How does that work?

Prompt: I have a local program that needs access to a MySQL database. How can I set up a Docker container accordingly?

ChatGPT provides a detailed seven-point description, starting with the Docker installation and ending with a Python application example. We've only removed the three central Docker commands here:

```
docker network create my_network
docker pull mysql:latest
docker run --name my_mysql_container --network my_network \
  -e MYSQL_ROOT_PASSWORD=rootpassword \
  -e MYSQL_DATABASE=mydatabase \
  -e MYSQL_USER=myuser \
  -e MYSQL_PASSWORD=mypassword \
  -p 3306:3306 -d mysql:latest
```

Basically, it works. Creating a separate network for the container is completely superfluous here, but doesn't interfere. In the last point, ChatGPT points out that all data stored in the database will be lost as soon as the container terminates. To prevent this, you must create a *volume* when creating the container. You can reuse this volume later and transfer the database from one container to the next.

> If you want to persist data even after the container stops, you can mount a volume. Modify the Docker run command to include a volume:
>
> ```
> docker run ... -v my_mysql_data:/var/lib/mysql
> ```

ChatGPT will tell you how to transfer the data to a new container if required in a follow-up question. We've only quoted the central sentence of the answer here.

Prompt: After ending the first container, how would I create a new container reusing the volume?

> When creating the new container, use the same volume (`my_mysql_data`) . . .

8.4.3 Proxy Server

Modern web applications and REST APIs are usually connected to the internet via a proxy server. AI tools can explain how they work and how to set them up:

Prompt: I want to build a REST application using Python and Django. The application will run in a Docker container. Do I need a proxy server in this case to make my application visible to the internet?

ChatGPT first explains how a reverse proxy server works and its advantages and then provides instructions for a corresponding Docker setup. If you do something like this for the first time, you'll probably only get there with a few additional questions. However, if you already have some experience in this regard, the `Dockerfile` and `docker-compose.yml` files compiled by ChatGPT are an excellent starting point.

> In the context of deploying a REST application using Python and Django in a Docker container, whether you need a proxy server depends on your de-

> ployment architecture and requirements. Here's a breakdown to help you understand when and why you might need a proxy server. [more details]
>
> Here's a high-level overview of setting up a Django REST application with Docker and Nginx as a reverse proxy:
>
> - Dockerize Your Django Application [more details]
> - Create a Dockerfile [more details]
> - Create a `docker-compose.yml` [more details]
> - Set Up Nginx as a Reverse Proxy [more details]
> - Run Your Application

ChatGPT is currently only able to visualize the data in a rather dull ASCII representation. Here, GPT-4o doesn't yet live up to the "omni" promises.

Prompt: Can you visualize the setup in a diagram? Please use ASCII art.

> Certainly! Here's a textual description of the diagram for the setup described:

```
+----------------------------+
|      Client Browser        |
+-----------+----------------+
            |
            v
+-----------+----------------+
|       Nginx Reverse Proxy  |
|  (Handles SSL, Load        |
|   Balancing, etc.)         |
+-----------+----------------+
            |
            v
+-----------+-----------------+
| Docker Container Network    |
|                             |
|  +-----------+   +---------+|
|  |  Web      |   |  DB     ||
|  | Container|   |Container||
|  +-----------+   +---------+|
```

> ```
> |    (Django)        (Postgres)|
> +----------------------------+
> ```

8.4.4 Fixing Network Issues in a Virtual Machine with a Web Server

You've installed Linux, a web server, and various other tools in a virtual machine. But you fail to access the web server from your host. Can ChatGPT help?

Remember to be as precise as possible. Which virtualization program? Which Linux distribution? Which web server? Which host operating system? What error message? What have you already tried to narrow down the error?

Prompt: I am using VirtualBox on my Windows notebook. I have set up a virtual machine with Fedora Linux. Within Fedora, I have installed Apache.

- `systemctl` status `httpd` says it is running fine.
- hostname `-I` reveals that my IP address is 10.0.2.15.

However, on my host system, I can't access the website. Whenever I try to open *http://10.0.2.15*, the web browser displays `ERR_CONNECTION_TIMED_OUT`. What could cause this error?

> The issue you're experiencing is likely due to the network configuration of your VirtualBox setup. By default, VirtualBox often uses NAT (Network Address Translation) for networking, which can cause issues with accessing the guest machine from the host. Here are a few steps to resolve this:
>
> - Change Network Adapter to Bridged Adapter [more details]
> - Determine New IP Address [more details]
> - Update Firewall Rules. Ensure that the firewall on Fedora allows HTTP traffic. First, check the status of the firewall:
>
>   ```
>   sudo firewall-cmd --state
>   ```
>
>   If the firewall is running, allow HTTP traffic:

> ```
> sudo firewall-cmd --permanent --add-service=http
> sudo firewall-cmd --reload
> ```
>
> - Access the Web Server [more details]

In our experience, ChatGPT's response is perfect. The AI tool points out the two most common causes of errors. First, the default network configuration of VirtualBox is unsuitable. You need to switch to *bridged networking*. Second, a firewall may be active in the virtual machine. (This is definitely the case with Fedora.) You must define an exception rule that allows network traffic for HTTP.

If you find it difficult to follow the instructions, you can ask follow-up questions. In the interplay between host system, virtualization program, and virtual machine, it's not always easy to identify the correct location for the individual configuration steps.

Prompt: Should I change the firewall settings on my Windows notebook or within the virtual machine?

> To ensure that your Fedora Apache web server is accessible from your Windows host, you should primarily focus on the firewall settings within the virtual machine (Fedora). [more details]

8.4.5 Linux Server Setup

You want to set up a Linux server. You use Redundant Array of Independent Disks 1 (RAID-1), that is, two solid-state drives (SSDs) whose contents are mirrored. If one of the two volumes crashes, you don't have to worry about losing any data.

Now the boot files are usually only on the first disk. If that disk fails, you'll not lose any data, but you'll no longer be able to boot your computer. This can be solved by correctly configuring the GRUB boot loader on *both* disks. But how does that work?

Prompt: Linux: I want to set up a server with two disks using RAID-1. The server should be bootable no matter whether the first or the second disk fails. Can you help?

ChatGPT provides a detailed instruction that starts with partitioning the disks, then explains the RAID setup, and finally covers the installation of GRUB. We've only reproduced the crucial point for the question here:

> Ensure GRUB is installed on both disks. You can do this manually after the installation:
>
> ```
> sudo grub-install /dev/sda
> sudo grub-install /dev/sdb
> sudo update-grub
> ```

It's therefore important that the GRUB boot loader is installed on *both* disks, which typically isn't the case. Please note that you must use the device names `dev/nvme0n1` and `/dev/nvme1n1` instead of `/dev/sda` and `/dev/sdb` for modern servers with Non-Volatile Memory Express (NVMe) SSDs.

8.4.6 Additional Prompts

We want to round off this section with a few prompt examples without going into the results in detail.

Prompt: Bash, `find` command: Is there a way to search for files within a certain date range?

Prompt: PowerShell: How do I list files sorted by modified date?

Prompt: Linux: I want to set up a shared directory for five students to work together. By default, each student should be able to read and write files created by other students in the group. How do I do this?

Prompt: Linux: I have a Logical Volume Management (LVM) setup. I need to increase the size of one of my logical volumes, but the volume group is already fully used. Therefore, I plan to add a second disk. How can I add this disk in my LVM system? How do I increase the size of the logical volume and the file system on it?

Prompt: Linux Server with Postfix mail server: I can send but I can't receive mail. Can you help me debug this?

Prompt: Linux Server with nginx web server. How can I set up a Let's Encrypt certificate for the main domain and three subdomains (**example.com**, **www.example.com**, **help.example.com**, and **support.example.com**).

Prompt: Ubuntu server: I would like to set up a firewall that blocks all incoming traffic with these exceptions: ping, SSH, HTTP, and HTTPS. How should I proceed?

Prompt: ReadyMedia: Can I use ReadyMedia for radio streams (`.m3u`)? If not, what are my options for a simple setup on the Raspberry Pi?

Prompt: Linux: I want to implement geo-blocking with a nftables firewall to protect my services against attacks from certain countries. What are my options?